Practice problems #22

1. Find A such that the given set is Col A:

$$\left\{ \begin{bmatrix} b-c \\ 2b+c+d \\ 5c-4d \\ d \end{bmatrix} : b, c, d \,\text{real} \right\}.$$

2. Let $T : V \to W$ be a linear transformation from a vector space V into a vector space W. Prove that Range T is a subspace of W.

3. Define a linear transformation $T : \mathbf{P_2} \to \mathbf{R^2}$ by $T(p) = \begin{bmatrix} p(0) \\ p(1) \end{bmatrix}$. For instance, if $p(t) = 3 + 5t + 7t^2$, then $T(p) = \begin{bmatrix} 3 \\ 15 \end{bmatrix}$.

   (a) Show that T is a linear transformation.

   (b) Find a polynomial $p \in \mathbf{P_2}$ that spans the Kernel of T, i.e., find a spanning set for Ker T.

   (c) Describe the range of T.

Answers are on page 2.

#1: $\begin{bmatrix} 1 & -1 & 0 \\ 2 & 1 & 1 \\ 0 & 5 & -4 \\ 0 & 0 & 1 \end{bmatrix}$

#2: Note that Range $T = \{b \in W : T(x) = b \text{ for some } x \in V\}$.
Range $T \subseteq W$ is clear.
The zero vector of W is in range of T because $T(0) = 0$ (since T is linear).
If b_1, b_2 are in Range T (which means there are x_1 and x_2 in V such that $T(x_1) = b_1$ and $T(x_2) = b_2$), then $b_1 + b_2 \in$ Range T because $T(x) = b_1 + b_2$ has a solution (what's the solution?)
Show similarly that if $b \in$ Range T and c is any scalar, then $T(x) = cb$ has a solution (what's the solution?) so that $cb \in$ Range T.

#3(a): Let p, q be polynomials in $\mathbf{P_2}$, then

$$T(p+q) = \begin{bmatrix} (p+q)(0) \\ (p+q)(1) \end{bmatrix} = \begin{bmatrix} p(0) + q(0) \\ p(1) + q(1) \end{bmatrix} = \begin{bmatrix} p(0) \\ p(1) \end{bmatrix} + \begin{bmatrix} q(0) \\ q(1) \end{bmatrix} = T(p) + T(q).$$

Similarly, show that $T(cp) = cT(p)$.

#3(b): Ker $T = \{p \in \mathbf{P_2} : p(0) = 0 \text{ and } p(1) = 0\} = \{\alpha t(t-1) : \alpha \in \mathbf{R}\}$, i.e.,
Ker $T =$ Span $\{t(t-1)\}$. So, the spanning set for Ker T is $\{t(t-1)\}$.

#3(c): Given any vector $\begin{bmatrix} \alpha \\ \beta \end{bmatrix} \in \mathbf{R^2}$, we can form a polynomial p so that $p(0) = \alpha$ and $p(1) = \beta$. For instance, $p(t) = (\beta - \alpha)t + \alpha$ is such polynomial.
Hence Range $T = \mathbf{R^2}$.